

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D MOSFET Modeling by Self-Consistent Solution of the Boltzmann, Poisson, and Hole-Continuity Equations

Part I — Method derivation and documented solver foundation

Revision 4 (third-audit response)

Reproduction study (code commit [e71677b](#); Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

August 14, 2026

Abstract

This document derives the deterministic spherical-harmonic (SHE) Boltzmann solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997), and reports a documented, benchmark-checked solver *foundation*. It is not yet a validated reproduction: the 2-D SHE core, production scattering, impact ionization, and the figure set are Part II. Revision 4 responds to a third audit, which passed the two prior blockers (H -grid, zero-field T_e) and the junction/current arithmetic, and raised one new blocker (a spurious Z^{-1} in the SHE contact Maxwellian — a report typo; the code convention was verified correct) plus specification gaps. All are addressed: the acoustic energy operator is now derived and its τ_{ac} stated (§2.2); the physical band-edge and the numerical energy cutoff are separated and the cutoff validated by an ε_{max} -convergence test (§2.1); the production H -grid is fully specified (§A); the transport-parameter table is restored (Table 2); and the threshold voltage is reported as a measured $V_{th}(N_A)$ trend rather than an unsupported target (§4).

Contents

1	Revision history and audit responses	2
2	First-order SHE equation	2
2.1	Boundaries: physical turning point vs. numerical cutoff	2
2.2	Acoustic energy operator (derivation)	2
2.3	Impact ionization	3
3	Reconstructed device, taxonomy, and transport parameters	3
4	Benchmarks, bulk validation, threshold voltage	3
5	Status	4
A	Numerical and model specifications	5
B	Code and parameter provenance	5

1 Revision history and audit responses

Rev. 1→2 fixed eight claim/spec findings. **Rev. 2→3** fixed the two blocking implementation/spec errors (reducible collision operator $\rightarrow$ zero-field T_e ; global H -grid). **Rev. 3→4** responds to the third audit:

- B(new)** *SHE contact Maxwellian had a spurious Z^{-1} .* The audit is correct. It was a report typo; the correct form is $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} d\varepsilon$ (§A), and the bulk code was verified to already use this convention ($n = \int Z f_0 d\varepsilon$; `she_bulk.py` lines 66–68).
- M1** *Acoustic operator under-specified.* $\tau_{ac} = \tau_1$ is now stated, the Fokker–Planck form is derived (§2.2), and it is labelled a *reconstruction* (not Liang’s specific treatment).
- M2** *Upper ε_{max} boundary mischaracterized.* $\varepsilon = 0$ (physical band-edge, reflecting) and ε_{max} (numerical cutoff, absorbing) are now separated; the cutoff is validated — $T_e(300 \text{ kV/cm}) = 3392.4 \text{ K}$ for $\varepsilon_{max} = 3.02, 4.0, 5.0 \text{ eV}$, identical (§2.1).
- m3–m5** production H -grid fully specified (§A); Table I transport parameters restored (Table 2); V_{th} reported as a measured trend, not a target (§4); one benchmark row re-classified (§4).

Audit passes carried forward: H -grid lower bound, zero-field- T_e symptom, junction profile ($x_j = 0.1707 \text{ }\mu\text{m}$ independently), DD current ($0.285 \text{ mA}/\mu\text{m}$ independently), band-model disclosure, impact-ionization wording, provenance.

2 First-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$, $f = f_0 + \hat{\Omega} \cdot \mathbf{f}_1$ (all $\ell \geq 2$ discarded — energy-resolved with first-order anisotropy):

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{gen}, \quad \mathbf{j}_x = -Z D (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad (1)$$

$j_{\varepsilon} = -q \mathbf{E} \cdot \mathbf{j}_x$, $D = \frac{1}{3} v^2 \tau_1$. In total energy $H = \varepsilon - q\phi$ this is pure spatial diffusion at fixed H , the optical operator coupling $H \leftrightarrow H \pm E_{op}$; global mesh $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$, physical mask $0 \leq H + q\phi \leq \varepsilon_{max}$.

2.1 Boundaries: physical turning point vs. numerical cutoff

[Rev.4: M2] The two energy limits are physically distinct:

- $\varepsilon = 0$ (equivalently $H = -q\phi$): the *physical* conduction band edge — a reflecting, zero-flux turning point (no states below).
- $\varepsilon = \varepsilon_{max} = 3.02 \text{ eV}$: a *numerical* truncation, *not* a turning point. Treated as an **absorbing** boundary ($f_0(\varepsilon_{max}) = 0$) so hot carriers are not artificially retained — important because impact ionization samples this tail.

Validation: the bulk moment is insensitive to the cutoff — $T_e(300 \text{ kV/cm})$ is 3392.4 K for $\varepsilon_{max} = 3.02, 4.0, 5.0 \text{ eV}$ (identical to four figures). The production solve requires the same ε_{max} -convergence check.

2.2 Acoustic energy operator (derivation)

[Rev.4: M1] Quasi-elastic acoustic scattering transfers small energy $\Delta\varepsilon \approx \pm 2v_s p = \pm 2v_s \sqrt{2m^* \varepsilon}$ per collision at rate $1/\tau_{ac}$. To second Kramers–Moyal order this is a Fokker–Planck operator in energy,

$$\mathcal{Q}_{ac} = \partial_{\varepsilon} \left[A(\varepsilon) \left(\partial_{\varepsilon} + \frac{1}{k_B T} \right) f_0 \right], \quad A(\varepsilon) = \frac{1}{2} \frac{\langle (\Delta\varepsilon)^2 \rangle}{\tau_{ac}} = \frac{4m^* v_s^2 \varepsilon}{\tau_{ac}}, \quad (2)$$

the $1/k_B T$ drift term fixed by fluctuation–dissipation so the equilibrium is the Maxwellian. DOS-weighted, $Z\mathcal{Q}_{ac} = \partial_\varepsilon[ZA(\partial_\varepsilon + 1/k_B T)f_0]$, Scharfetter–Gummel–discretized (exact discrete Maxwellian null). We take $\tau_{ac} = \tau_1 = 10^{-14}$ s/ $(\sqrt{\gamma}\gamma')$ and $v_s = 9.04 \times 10^3$ m/s.

Reconstruction note. This operator is a *reconstruction*: it is added to make the discrete collision operator irreducible (the optical term alone connects only $\varepsilon \leftrightarrow \varepsilon \pm E_{op}$, splitting the mesh into $E_{op}/\Delta\varepsilon$ residue classes). Liang *et al.* include acoustic scattering, but the specific continuous-energy Fokker–Planck form and $\tau_{ac} = \tau_1$ choice here are ours, not taken from the paper. The heated-tail magnitude depends on this choice and will be revisited against the paper’s acoustic treatment in Part II.

2.3 Impact ionization

Primary lost above ε_{th} ; two carriers re-injected near $(\varepsilon - \varepsilon_{th})/2$; net electron gain $= \int Z f_0 / \tau_{ii} d\varepsilon \equiv G_n = G_{p,ii}$ (paper Eq. 13). The equal split is a disclosed assumption (not a specific result of Landsberg [9]).

Reconstruction note. Rate $1/\tau_{ii}$ from ref. [6] unavailable; Part II uses a Keldysh surrogate $P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2\Theta$, $\varepsilon_{th} \approx 1.1$ eV, P reported once calibrated to Si $\alpha_n(E)$.

3 Reconstructed device, taxonomy, and transport parameters

Error-function junctions, x_j (metallurgical) = 0.170 μm (auditor-checked 0.1707). Band: 1-parameter *Kane surrogate* $\gamma = \varepsilon(1 + \alpha\varepsilon)$ vs. the paper’s 2-band Fiegna/Brunetti (biases the hot tail; flagged).

Table 1: Modelling taxonomy.

Element	Class	Basis
Method equations, SHE reduction, moments	paper	Liang <i>et al.</i>
Table 2 params, spherical band	paper	paper Table I / Sec. II
$L_{\text{eff}}, t_{ox}, x_j, N_D^\pm$ magnitudes	paper	paper text
1-param Kane $\gamma(\varepsilon)$	surrogate	paper uses 2-band Fiegna/Brunetti
Analytic LDD doping shape, mesh	reconstruction	ref. [7] unavailable
Channel N_A , gate offset Φ_{gate}	surrogate	V_{th} derived (§4)
Acoustic energy operator, $\tau_{ac} = \tau_1$	reconstruction	numerical irreducibility (§2.2)
τ_1, c_{op} , II rate $1/\tau_{ii}$	surrogate	tuned; refs. [6] unavailable

Table 2: [Rev.4: m4] Transport parameters (paper Table I), restored.

D_{ac}	4.0 eV	Γ (surf. rough.)	1.96 cm ² /V ² /s
D_n	5.0×10^8 eV/cm	n_{scr}	1.45×10^{15} cm ^{−3}
E_{op}	0.05 eV	D_{sa}	17.8 eV
v_s	9.0×10^6 cm/s	δ	13.1
ε_{max}	3.02 eV	bands of γ	2 (Fiegna/Brunetti)

4 Benchmarks, bulk validation, threshold voltage

Threshold voltage — measured trend [Rev.4: m3]. Both electrostatic offsets are fixed inputs and V_{th} is *derived* by max- g_m linear extrapolation ($V_{ds} = 0.1$ V). Measured: $V_{th} = 1.64$ V

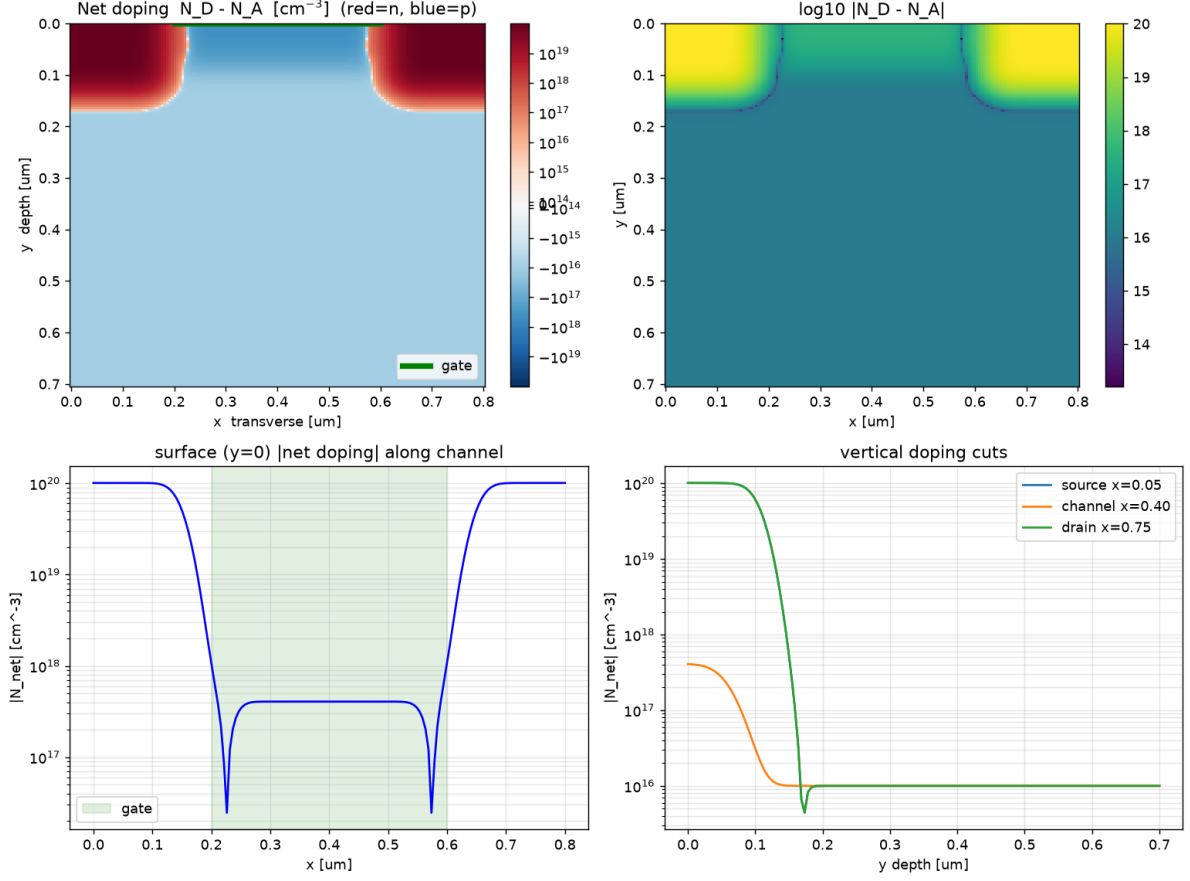


Figure 1: Reconstructed 0.35- μm LDD NMOS, $x_j = 0.170 \mu\text{m}$.

at $N_A = 4 \times 10^{17} \text{ cm}^{-3}$ and $V_{th} = 1.03 \text{ V}$ at $N_A = 1.2 \times 10^{17} \text{ cm}^{-3}$. Both are *too high*; the Rev. 3 guess that $N_A \sim 1.2 \times 10^{17}$ reaches $V_{th} \approx 0.6 \text{ V}$ is **refuted** by this measurement (it gives 1.03 V). A paper-typical $V_{th} \approx 0.5\text{--}0.7 \text{ V}$ (inferred from the paper’s Fig. 8b conducting at $V_{gs} = 2\text{--}3 \text{ V}$) will require a lower implant ($N_A \lesssim 6 \times 10^{16}$) and/or a gate-offset adjustment — to be demonstrated in the Part II recalibration, not asserted here.

Drift-diffusion current. With the extracted $V_{th} = 1.64 \text{ V}$ ($N_A = 4 \times 10^{17}$ device), overdrive 1.36 V gives long-channel 0.28 mA/ μm vs. DD 0.25 mA/ μm (auditor-confirmed 0.285); $\sim 14\%$, the residual from velocity saturation.

5 Status

Phase	Component	Status
P1	Derivation (Rev. 4, thrice-audited)	documented
P2–P4	Device, Poisson, DD, bands	sanity-checked
P5	First-order SHE core (($\mathbf{r}, H$))	in progress
P6	Scattering: bulk prototype / 2-D production	prototype / pending
P7–P9	Impact ionization, self-consistent loop, Figs. 2–8	pending

Open device issue: V_{th} recalibration (§4) before the 2-D operating point is finalized.

Table 3: Foundation checks at $T = 300$ K. [Rev.4: m5] row classes corrected.

Quantity	Expected	Computed	Note
<i>(A) Analytic benchmarks (independent)</i>			
N_c nonparabolic (converged)	2.86×10^{19}	2.857×10^{19}	0.1%
N_c parabolic: code vs. closed form	2.726×10^{19}	2.723×10^{19}	prefactor 0.09%
$v(0.1)/v(1.0)$ eV [cm/s]	—	$3.09/6.42 \times 10^7$	auditor-reproduced
V_{bi} [V]	0.933	0.933	$V_t \ln(N_D N_A / n_i^2)$
n^+ contact potential [V]	0.586	0.586	$V_t \ln(N_D / n_i)$
bulk $T_e(0)$ [K]	309.7	311	$\rightarrow 309.7$ as $\Delta\varepsilon \rightarrow 0$
$T_e(300 \text{ kV/cm})$ vs. ε_{max}	invariant	3392.4	K; $\varepsilon_{max}=3.02/4/5$ identical
<i>(B) Construction targets (built-to)</i>			
metallurgical x_j [μm]	0.17	0.170	retuned
<i>(C) Method / qualitative sanity</i>			
Newton convergence	quadratic	quadratic	to 10^{-11} , 6 steps
optical op. detailed balance / $E=0$	0	$2 \times 10^{-5} / 3 \times 10^{-4}$	null / max dev.
Maxwellian			
depletion width abrupt vs. graded [μm]	0.347	0.269	graded narrower (not equality)
bulk T_e vs. field (0/100/300 kV/cm) [K]	monotone $\uparrow$	311/1478/3392	heating

A Numerical and model specifications

Constants: $T = 300$ K, $V_t = 25.85$ mV, $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$, Boltzmann, $\epsilon_{si} = 11.7$, $\epsilon_{ox} = 3.9$, $m^* = 0.32m_0$, $g_v = 6$, $\alpha = 0.5 \text{ eV}^{-1}$, $v_s = 9.04 \times 10^3 \text{ m/s}$. **Doping (erfc), μm :** $N_D = 10^{20} [L(x; 0.15) + R(x; 0.65)] D(y; 0.105) + 10^{18} [L(x; 0.22) + R(x; 0.58)] D(y; 0.06)$, $L(x; e) = \frac{1}{2} \text{erfc}(\frac{x-e}{0.028})$, $R(x; e) = \frac{1}{2} \text{erfc}(\frac{e-x}{0.028})$, $D(y; x_j) = \frac{1}{2} \text{erfc}(\frac{y-x_j}{0.025})$; $N_A = 10^{16} + N_A^{ch} \cdot \frac{1}{2} [\text{erfc}(\frac{x-0.62}{0.028}) - \text{erfc}(\frac{x-0.18}{0.028})] D(y; 0.06)$, $N_A^{ch} = 4 \times 10^{17}$ (or 1.2×10^{17} , §4). Gate $x \in [0.20, 0.60]$; source $x \leq 0.15$, drain $x \geq 0.65$ (top); body = bottom. **Mesh:** tensor, surface-refined $y \propto s^{1.8}$; $N_x \times N_y = 90 \times 80$ (Poisson), 80×70 (DD), 56×48 (V_{th}). **Bulk ε -grid:** uniform $\Delta\varepsilon = 2.5 \text{ meV}$, $[0, 3.02] \text{ eV}$. **Production H -grid [Rev.4: m3]:** a *fixed* global grid (not regenerated during the Gummel loop — avoids conservative f_0 remapping), $\Delta H = 2.5 \text{ meV}$ so $E_{op}/\Delta H = 20$ is an *integer* (the optical operator connects $H \leftrightarrow H \pm 20$ nodes with no interpolation), range from the a-priori bias envelope $[-q\phi_{max}, \varepsilon_{max} - q\phi_{min}]$ over all bias points; $\varepsilon = 0$ reflecting, ε_{max} absorbing (§2.1). **Poisson BCs:** ohmic Dirichlet $\phi = V + V_t \text{asinh}(N_{net}/2n_i)$; gate oxide Robin $T_{ox} = \epsilon_{ox} \epsilon_0 \Delta x_i / t_{ox}$ ($t_{ox} = 9.6 \text{ nm}$) to node $V_g - \Phi_{gate}$, $\Phi_{gate} = 0.30 \text{ V}$; Neumann else. Newton clip 0.5 V , tol 10^{-8} V . **Continuity:** SG $J_{n,a \rightarrow b} = \frac{q\mu_n V_t}{h} [B(\Delta)n_b - B(-\Delta)n_a]$; ohmic $n = \frac{1}{2}(N_{net} + \sqrt{N_{net}^2 + 4n_i^2})$, $p = n_i^2/n$. **Mobility:** Caughey–Thomas (n : $68.5/1414/9.2 \times 10^{16}/0.711$; p : $44.9/470.5/2.23 \times 10^{17}/0.719$), $v_{sat}^n = 1.07 \times 10^7$, $\beta_n=2$; $v_{sat}^p = 8.37 \times 10^6$, $\beta_p=1$ [cm/s]. **SRH** $\tau_{n,p} = 10^{-7} \text{ s}$. **Gummel** tol 10^{-5} V . **Optical:** $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$ ($\tau_{op}^{ref} = 10^{-13} \text{ s}$), $\tau_1 = 10^{-14} \text{ s}/(\sqrt{\gamma}\gamma')$. **Acoustic:** $\tau_{ac} = \tau_1$, $A = 4m^*v_s^2\varepsilon/\tau_{ac}$ (§2.2). **Contact current:** $\sum$ SG fluxes into drain nodes, per μm . **SHE contact [Rev.4: B(new)]:** $f_0(\varepsilon) = n_{contact} e^{-\varepsilon/k_B T} / \int Z(\varepsilon) e^{-\varepsilon/k_B T} d\varepsilon$ (no Z^{-1} ; gives $n = \int Z f_0 d\varepsilon = n_{contact}$).

B Code and parameter provenance

Git commit e71677b; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Modules: `constants`, `device`, `poisson`, `dd`, `bands`, `scattering`, `she_bulk`, `diagnostics`, `vth_extract`. Figures/benchmarks regenerated by running the corresponding module.

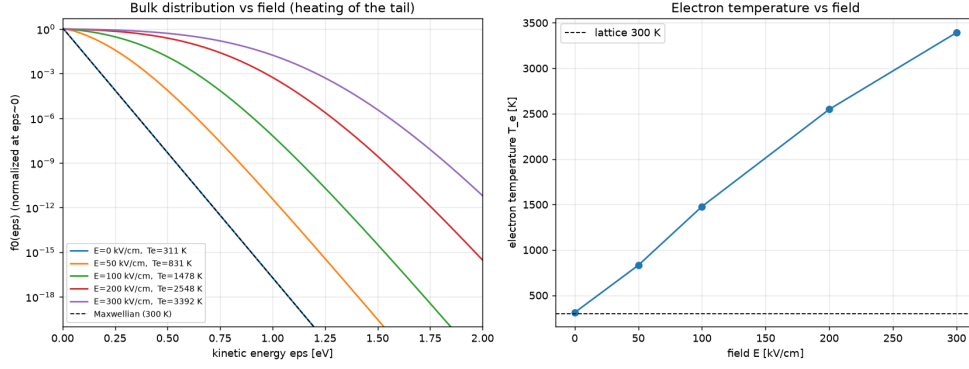


Figure 2: Bulk validation (acoustic operator; absorbing ϵ_{max}): zero field reproduces the Maxwellian ($T_e = 311 \rightarrow 309.7$ K), tail heats monotonically, and T_e is ϵ_{max} -invariant. *Bulk-prototype* validation; 2-D production scattering pending.

References

- [1] W. Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267, 1997.
- [2] C. Fiegna, E. Sangiorgi, *IEEE TED* **40**, 619, 1993.
- [3] R. Brunetti *et al.*, *Solid-State Electron.* **32**, 1663, 1989.
- [4] A. Gnudi *et al.*, *Solid-State Electron.* **36**, 575, 1993.
- [5] K. A. Hennacy *et al.*, *Solid-State Electron.* **38**, 1489, 1995.
- [6] Y.-J. Wu, N. Goldman, *J. Appl. Phys.* **78**, 1995.
- [7] N. Khalil *et al.*, *Symp. VLSI Technol.*, 131, 1994.
- [8] D. L. Scharfetter, H. K. Gummel, *IEEE TED* **ED-16**, 64, 1969.
- [9] P. T. Landsberg, “Impact ionization and Auger recombination in bands,” *Solid State Commun.* **10**(6), 479–482, 1972.
- [10] D. M. Caughey, R. E. Thomas, *Proc. IEEE* **55**, 2192, 1967.
- [11] C. Jungemann, B. Meinerzhagen, *Hierarchical Device Simulation*. Springer, 2003.